Two- and three-point objects

Plain lines expose connectivity. Scale=1 uses the compact baseline (0.8 times the former size). Rotated layouts are not extra graph topologies.

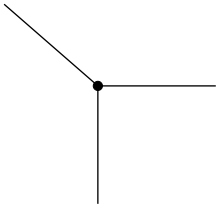
I01 Two-point object

`\TwoPointCorr`

Two external segments meet at one marked point.
A central point or optional blob, not a loop correction.

I02 Three-point: upper left

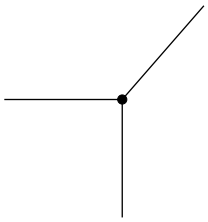
`\ThreePointCorr`



One trivalent vertex; logical slot 1 points upper left.
Four orientations retain the same three logical slots.

I03 Three-point: upper right

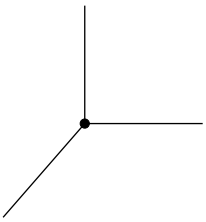
`\ThreePointCorr`



One trivalent vertex; logical slot 1 points upper right.
Four orientations retain the same three logical slots.

I04 Three-point: lower left

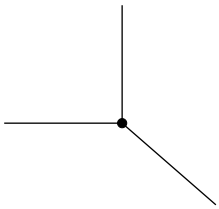
`\ThreePointCorr`



One trivalent vertex; logical slot 1 points lower left.
Four orientations retain the same three logical slots.

I05 Three-point: lower right

`\ThreePointCorr`



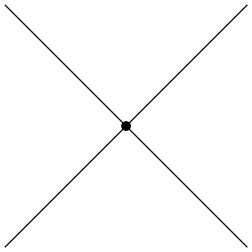
One trivalent vertex; logical slot 1 points lower right.
Four orientations retain the same three logical slots.

Contact and pairing layouts

Pairing presets are shown as implemented. A missing bridge is meaningful, not an omitted annotation.

I06 Central contact

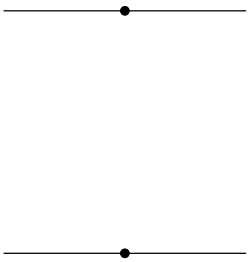
`\FourPointCorr`



Four external edges meet at one quartic vertex.
Connected four-point contact.

I07 Horizontal pairing

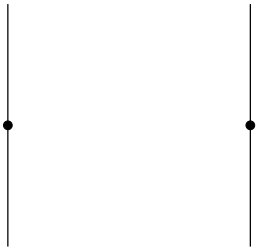
`\FourPointCorr`



Top pair and bottom pair each meet separately.
Disconnected pairs; there is no vertical exchange edge.

I08 Vertical pairing

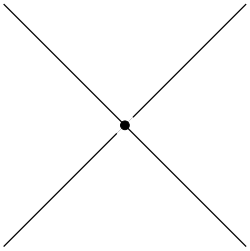
`\FourPointCorr`



Left pair and right pair each meet separately.
Disconnected pairs; there is no horizontal exchange edge.

I09 Crossed contact layout

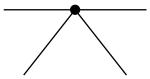
`\FourPointCorr`



Curved and straight external edges meet at a central point.
The central dot is an interaction, not an over/under crossing.

I10 Flat contact

`\FlatContactCorr`



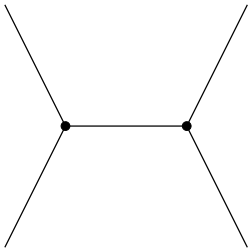
One quartic vertex with a flat upper pair.
A contact layout, not an additional loop topology.

Tree exchanges and four-legged bubbles

External slot order: 1 upper left, 2 lower left, 3 upper right, 4 lower right. No momentum annotations.

I11 s-channel tree

`\STreeChannelCorr`

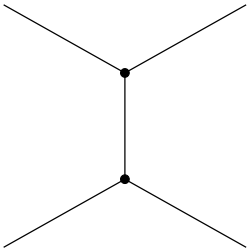


Two trivalent vertices joined by one exchange edge.

External insertions support S/T trees; U/internal hosts remain unsupported.

I12 t-channel tree

`\TTreeChannelCorr`

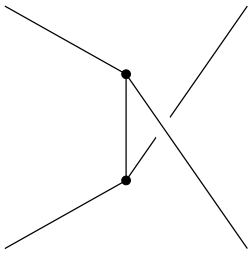


Two trivalent vertices joined by one exchange edge.

External insertions support S/T trees; U/internal hosts remain unsupported.

I13 u-channel tree

`\UTreeChannelCorr`

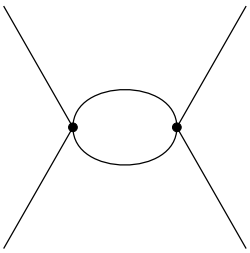


Two trivalent vertices joined by one exchange edge.

External insertions support S/T trees; U/internal hosts remain unsupported.

I14 s-channel bubble

`\SChannelCorr`

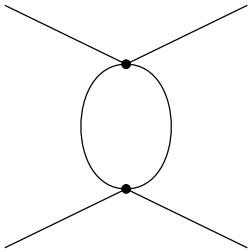


Two quartic vertices joined by two internal edges.

Four-legged bubble already exists; unlike a self-energy insertion.

I15 t-channel bubble

`\TChannelCorr`

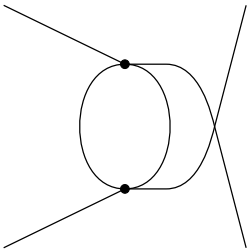


Two quartic vertices joined by two internal edges.

Four-legged bubble already exists; unlike a self-energy insertion.

I16 u-channel bubble

`\UChannelCorr`



Two quartic vertices joined by two internal edges.

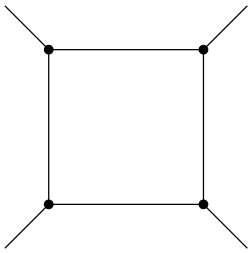
Four-legged bubble already exists; unlike a self-energy insertion.

Boxes and half-box layouts

The ordinary box already defaults to a square. Half-box presets include identity-specific marker variants.

I17 Square box

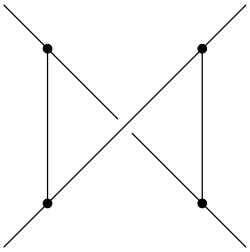
`\BoxLoopCorr`



Four trivalent vertices around a single loop.
Default inner spans are both 1.34; explicit overrides remain possible.

I18 Existing crossed box

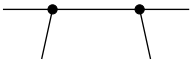
`\CrossBoxCorr`



Fixed left/right edges plus two crossing diagonals.
Default left/right fixed; use P02 for top/bottom fixed.

I19 Straight half-box

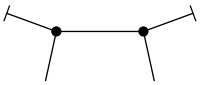
`\HalfBoxCorr`



Two trivalent vertices with a horizontal bridge.
An open tree helper, despite the half-box name.

I20 Angled half-propagator

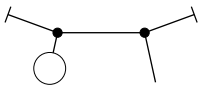
`\HalfBoxCorr`



Angled upper stubs on the same two-vertex graph.
A layout variant of the half-box, not a new loop.

I21 Left half-propagator marker

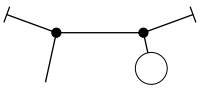
`\HalfBoxCorr`



Angled half-box with the left lower-leg circle preset.
Blank circle retained to show the preset; same underlying graph.

I22 Right half-propagator marker

`\HalfBoxCorr`



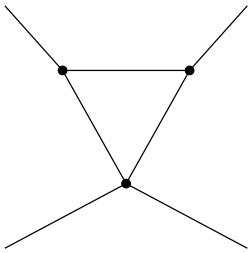
Angled half-box with the right lower-leg circle preset.
Blank circle retained to show the preset; same underlying graph.

Four triangle-contact orientations

Two trivalent vertices and one quartic vertex. Independent momentum placement is demonstrated on page 10.

I23 Quartic contact: bottom

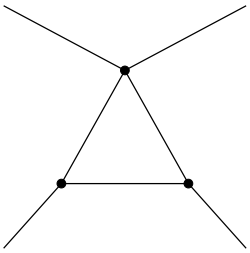
`\TriangleContactCorr`



The quartic/contact vertex lies on the bottom side.
Not the regular three-trivalent-vertex triangle attached to a tree.

I24 Quartic contact: top

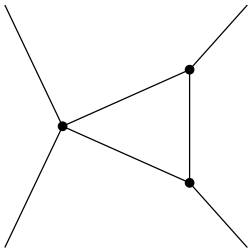
`\TriangleContactCorr`



The quartic/contact vertex lies on the top side.
Not the regular three-trivalent-vertex triangle attached to a tree.

I25 Quartic contact: left

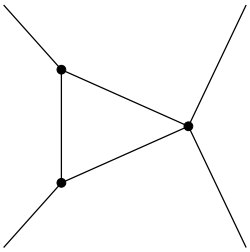
`\TriangleContactCorr`



The quartic/contact vertex lies on the left side.
Not the regular three-trivalent-vertex triangle attached to a tree.

I26 Quartic contact: right

`\TriangleContactCorr`



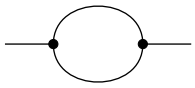
The quartic/contact vertex lies on the right side.
Not the regular three-trivalent-vertex triangle attached to a tree.

Loops and external bubble insertions

S/t external bubbles: all four legs (S page 12). External seagulls: all four legs (pages 8, 13-14).

I27 Self-energy bubble

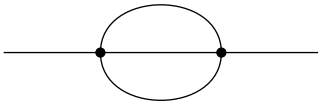
`\BubbleCorr`



Two trivalent vertices and two internal arcs; one loop.
Two external legs; distinct from a four-legged channel bubble.

I28 Sunset

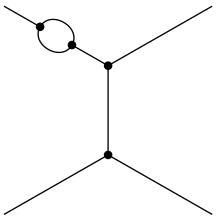
`\SunsetDiagram`



Two quartic vertices joined by three internal edges.
Two loops, not a one-loop self-energy bubble.

I29 Bubble on t-tree leg 1

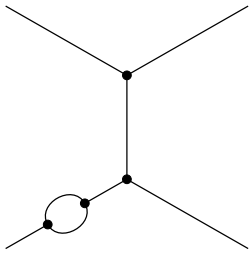
`\TTreeChannelCorr`



A self-energy bubble splits the upper-left external leg.
T-tree external slots 1-4; one self-energy insertion at a time.

B02 Bubble on t-tree leg 2

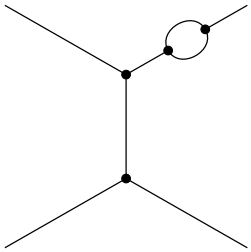
`\TTreeChannelCorr`



Two trivalent vertices split the lower-left external leg.
The other three legs and the exchange edge keep their logical identities.

B03 Bubble on t-tree leg 3

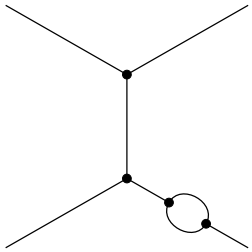
`\TTreeChannelCorr`



Two trivalent vertices split the upper-right external leg.
The other three legs and the exchange edge keep their logical identities.

B04 Bubble on t-tree leg 4

`\TTreeChannelCorr`



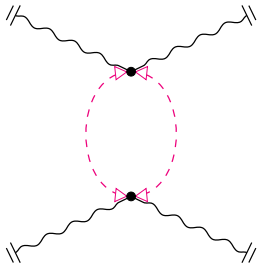
Two trivalent vertices split the lower-right external leg.
The other three legs and the exchange edge keep their logical identities.

Existing styles and annotations

Representative combinations, not a promise of every style on every topology. Recipes include all displayed options.

A01 Your pink Proca t-bubble

`\TChannelCorr`

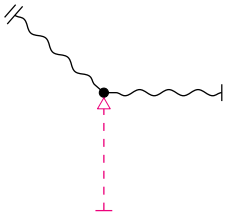


Four double-capped EW legs and two pink Proca internals.

Integrated endpoint checks cover visible contact and legacy overrides.

A02 Polarization and mixed legs

`\ThreePointCorr`

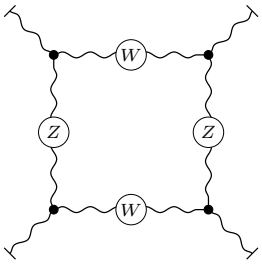


Double cap, single cap, and a pink scalar-polarized leg.

Scalar polarization is a style, not another topology.

A03 EW circles and single caps

`\BoxLoopCorr`

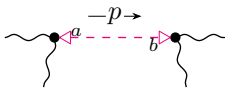


Independent internal particle circles and external endcaps.

Species labels decorate the existing square loop.

A04 Bridge momentum and indices

`\HalfBoxCorr`

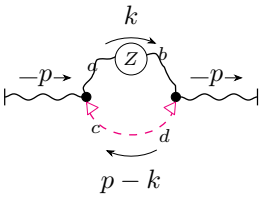


A pink bridge with one momentum and endpoint indices.

Omit global line=plain: it overrides the bridge-specific style.

A05 Curved momentum and indices

`\BubbleCorr`

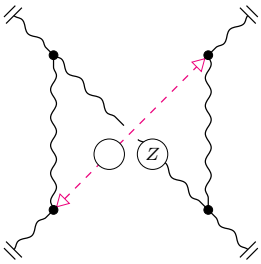


External flow, curved loop momenta, and four endpoint indices.

Representative annotation layout; crowded cases need dedicated checks.

A06 Crossing and Proca circles

`\CrossBoxCorr`



A blank circle on one pink diagonal and a Z on the other.

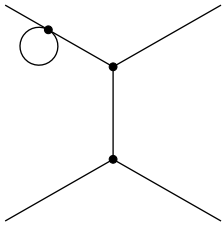
Circles shown deliberately; internal annotations are available separately.

Seagull insertion examples

Seagulls now support every external s/t-tree leg. These examples retain the original t-slot-1 styles; full placements: pages 13-14.

U01 Seagull: bare structure

`\TTreeChannelCorr`

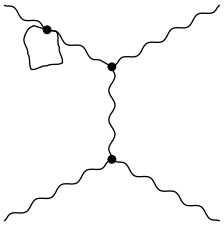


One quartic point on an external leg carries a self-loop.

Original t-slot-1 example; all s/t external slots 1-4 now supported.

U02 Seagull: EW line style

`\TTreeChannelCorr`

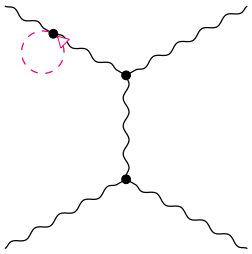


The same supported graph with wavy EW propagators.

Original t-slot-1 example; all s/t external slots 1-4 now supported.

S01 One endpoint marker

`\TTreeChannelCorr`

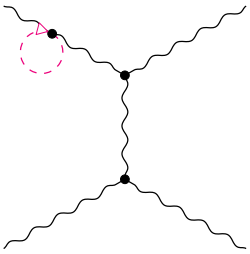


Select proca start on the pink seagull loop.

Loop endpoint selection follows the directed loop path; no caps or circles.

S02 Opposite endpoint marker

`\TTreeChannelCorr`

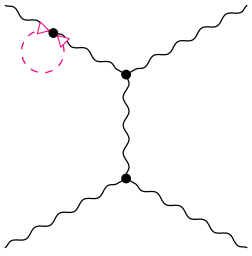


Select proca end on the pink seagull loop.

Loop endpoint selection follows the directed loop path; no caps or circles.

S03 Both endpoint markers

`\TTreeChannelCorr`



Select proca on the pink seagull loop.

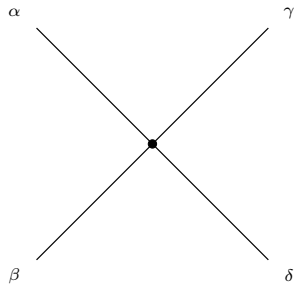
Loop endpoint selection follows the directed loop path; no caps or circles.

Integrated annotation controls

These controls came from the seagull branch. Examples show only the annotation feature being demonstrated.

F01 Compact external indices

`\FourPointCorr`

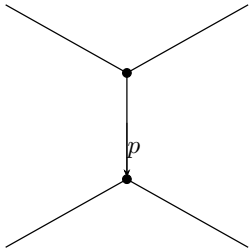


Compact indices on an empty field base.

Explicit leg labels take precedence over field/index formatting.

F02 Hide external momenta

`\TTreeChannelCorr`

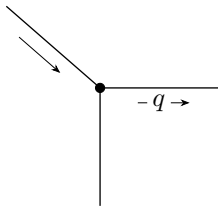


Only the exchange momentum remains; its side label leaves a continuous arrow.

Visibility acts on logical external slots and preserves internal flow.

F03 Blank label versus hidden arrow

`\ThreePointCorr`



Slot 1 has an unlabeled arrow, slot 2 has q, and slot 3 has no arrow.

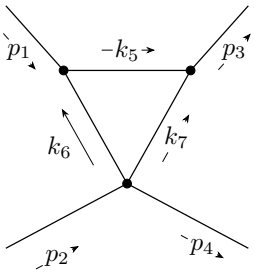
An empty entry hides text; external-momentum-hide hides the whole arrow.

Independent momentum placement

M02: seven logical slots; per-slot ranges, clearance, label positions and anchors. No new topology.

T01 Annotations: contact bottom

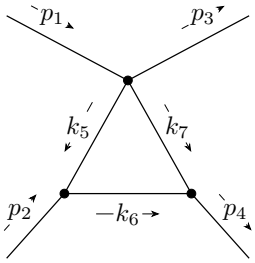
`\TriangleContactCorr`



Slot 2 arrow clearance and slot 6 side label are independent.
Compact quartic-side defaults sit nearer the outer endpoints.

T02 Annotations: contact top

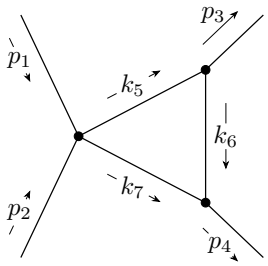
`\TriangleContactCorr`



Slot 5 has its own range; middle arrows face trivalent vertices.
Compact quartic-side defaults sit nearer the outer endpoints.

T03 Annotations: contact left

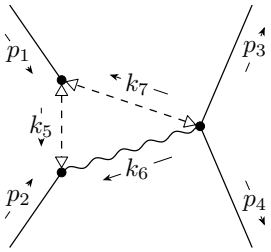
`\TriangleContactCorr`



Slot 3 uses a continuous arrow with its own side label.
Compact quartic-side defaults sit nearer the outer endpoints.

T04 Annotations: contact right

`\TriangleContactCorr`



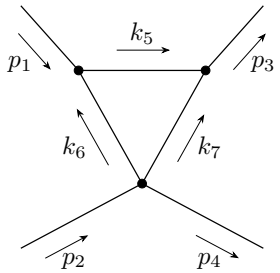
Unequal edges retain mixed styles and logical slots.
Compact quartic-side defaults sit nearer the outer endpoints.

Continuous arrows with outside labels

Choose triangle-contact-momentum-layout=side globally, or layout=side for an individual slot.

T05 Side labels: contact bottom

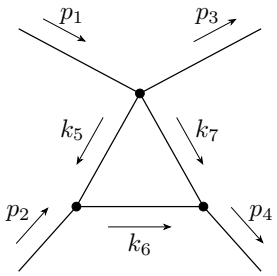
`\TriangleContactCorr`



Continuous arrows; labels sit outward from each edge.
Use layout=gap in an individual slot to mix both modes.

T06 Side labels: contact top

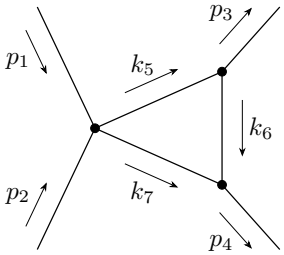
`\TriangleContactCorr`



Continuous arrows; labels sit outward from each edge.
Use layout=gap in an individual slot to mix both modes.

T07 Side labels: contact left

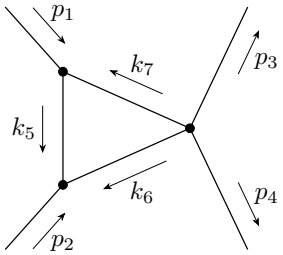
`\TriangleContactCorr`



Continuous arrows; labels sit outward from each edge.
Use layout=gap in an individual slot to mix both modes.

T08 Side labels: contact right

`\TriangleContactCorr`



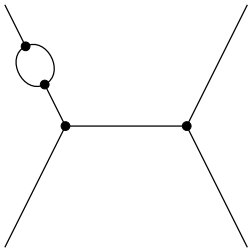
Continuous arrows; labels sit outward from each edge.
Use layout=gap in an individual slot to mix both modes.

External s-tree self-energy bubbles

One selected external edge is split by two trivalent vertices and two parallel arcs. Plain lines; no extra annotations.

C01 Bubble on s-tree leg 1

`\STreeChannelCorr`

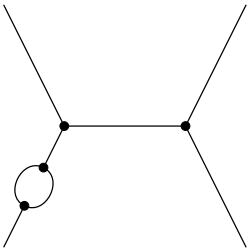


Self-energy correction on the upper-left external leg.

One insertion; internal-slot and u-tree support remain missing.

C02 Bubble on s-tree leg 2

`\STreeChannelCorr`

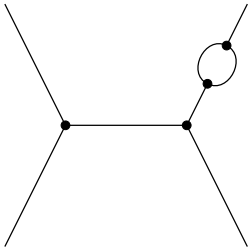


Self-energy correction on the lower-left external leg.

One insertion; internal-slot and u-tree support remain missing.

C03 Bubble on s-tree leg 3

`\STreeChannelCorr`

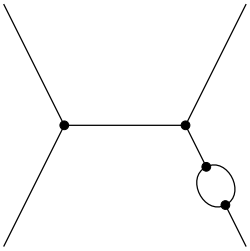


Self-energy correction on the upper-right external leg.

One insertion; internal-slot and u-tree support remain missing.

C04 Bubble on s-tree leg 4

`\STreeChannelCorr`



Self-energy correction on the lower-right external leg.

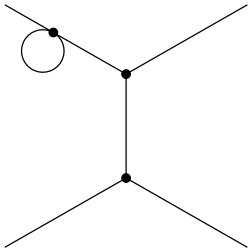
One insertion; internal-slot and u-tree support remain missing.

External t-tree seagulls

All t-tree seagull loops face inward by default. Explicit orientation controls remain available.

G01 Seagull on t-tree leg 1

`\TTreeChannelCorr`

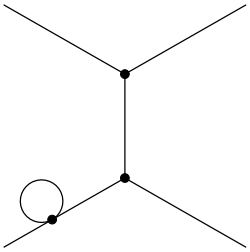


One quartic self-loop insertion on the upper-left leg.

Same graph as U01; repeated here to show the full four-leg matrix.

G02 Seagull on t-tree leg 2

`\TTreeChannelCorr`

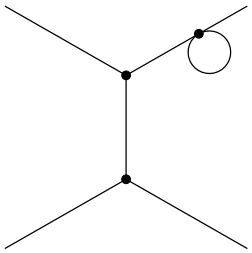


One quartic self-loop insertion on the lower-left leg.

External slots 1-4 only; one insertion.

G03 Seagull on t-tree leg 3

`\TTreeChannelCorr`

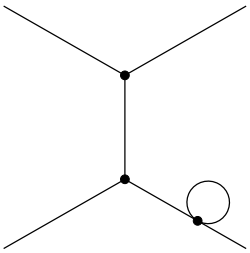


One quartic self-loop insertion on the upper-right leg.

External slots 1-4 only; one insertion.

G04 Seagull on t-tree leg 4

`\TTreeChannelCorr`



One quartic self-loop insertion on the lower-right leg.

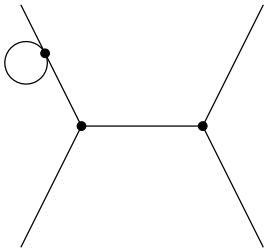
External slots 1-4 only; one insertion.

External s-tree seagulls

All s-tree seagull loops face outward by default. Each has one quartic insertion vertex and one self-loop.

G11 Seagull on s-tree leg 1

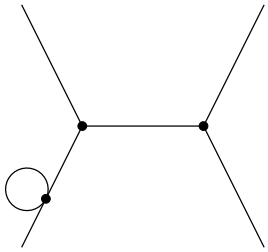
`\STreeChannelCorr`



One quartic self-loop insertion on the upper-left leg.
External slots 1-4 only; one insertion.

G12 Seagull on s-tree leg 2

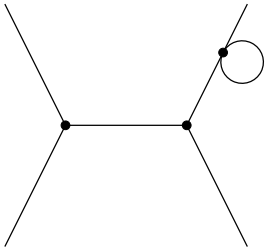
`\STreeChannelCorr`



One quartic self-loop insertion on the lower-left leg.
External slots 1-4 only; one insertion.

G13 Seagull on s-tree leg 3

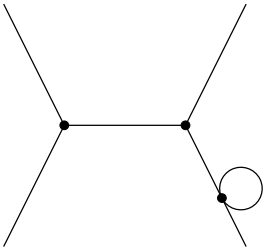
`\STreeChannelCorr`



One quartic self-loop insertion on the upper-right leg.
External slots 1-4 only; one insertion.

G14 Seagull on s-tree leg 4

`\STreeChannelCorr`



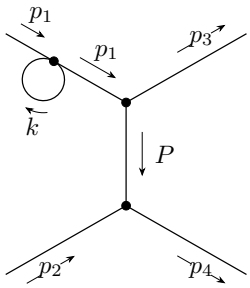
One quartic self-loop insertion on the lower-right leg.
External slots 1-4 only; one insertion.

Compact arrows and clear external labels

External momenta 1/2 enter and 3/4 leave. Labels clear their arrows and propagators; no caps, circles or indices.

H01 T-tree seagull on leg 1

`\TTreeChannelCorr`

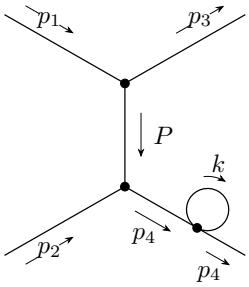


Inward T loop; compact host arrows.

Default outside labels; explicit placement keys still override.

H02 T-tree seagull on leg 4

`\TTreeChannelCorr`

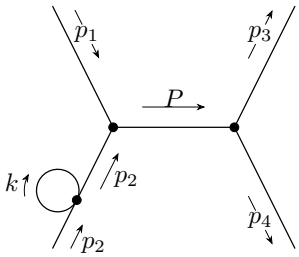


Inward T loop; compact host arrows.

Default outside labels; explicit placement keys still override.

H03 S-tree seagull on leg 2

`\STreeChannelCorr`

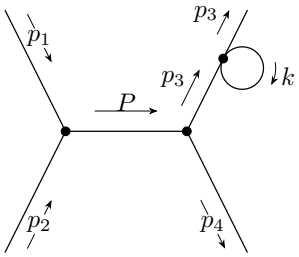


Outward S loop; compact host arrows.

Default outside labels; explicit placement keys still override.

H04 S-tree seagull on leg 3

`\STreeChannelCorr`



Outward S loop; compact host arrows.

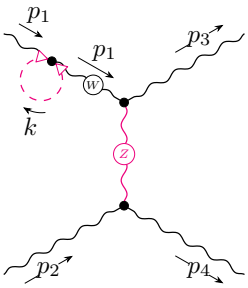
Default outside labels; explicit placement keys still override.

Pink Z stems, scalar-polarized loops and W host segments

No central momentum guides; pink Z stems and inner W circles retained. Pink loops: two endpoint arrows, no circles, k.

J01 T-tree seagull: pink Z stem

`\TTreeChannelCorr`

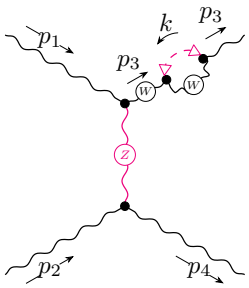


Pink Z stem; inner W; two-ended pink loop with k.

W centered in the visible arrow-to-vertex gap; central momentum hidden.

J02 T-tree bubble: pink Z stem

`\TTreeChannelCorr`

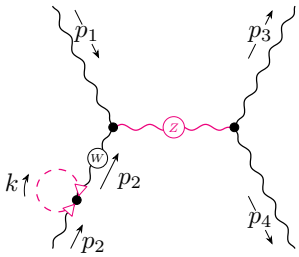


Pink Z stem; inner W; EW W arc and two-ended pink arc (k).

Central/W-loop guides hidden; no loop circles. Shorter W-loop wavelength.

J03 S-tree seagull: pink Z stem

`\STreeChannelCorr`

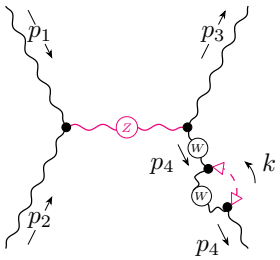


Pink Z stem; inner W; two-ended pink loop with k.

W centered in the visible arrow-to-vertex gap; central momentum hidden.

J04 S-tree bubble: pink Z stem

`\STreeChannelCorr`



Pink Z stem; inner W; EW W arc and two-ended pink arc (k).

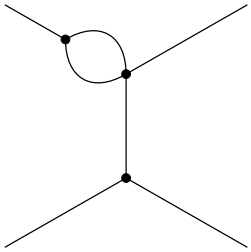
Central/W-loop guides hidden; inner-label gap 4pt. Shorter W-loop wavelength.

External t-tree swordfish diagrams

All four external slots. Plain lines and interaction vertices; one shared tree vertex, seven propagators, one loop.

K01 Swordfish on t-tree leg 1

`\TTreeChannelCorr`

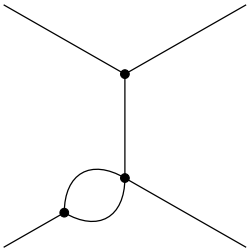


One trivalent loop vertex; two curves meet the actual quartic tree vertex.

One external insertion. No bridge or duplicate shared vertex.

K02 Swordfish on t-tree leg 2

`\TTreeChannelCorr`

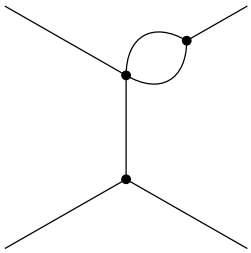


One trivalent loop vertex; two curves meet the actual quartic tree vertex.

One external insertion. No bridge or duplicate shared vertex.

K03 Swordfish on t-tree leg 3

`\TTreeChannelCorr`

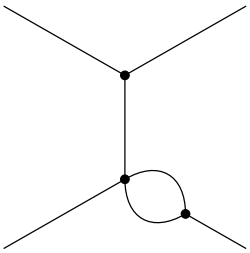


One trivalent loop vertex; two curves meet the actual quartic tree vertex.

One external insertion. No bridge or duplicate shared vertex.

K04 Swordfish on t-tree leg 4

`\TTreeChannelCorr`



One trivalent loop vertex; two curves meet the actual quartic tree vertex.

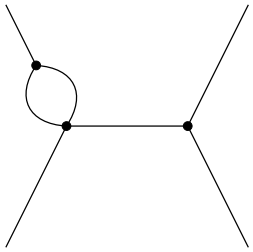
One external insertion. No bridge or duplicate shared vertex.

External s-tree swordfish diagrams

All four external slots. Plain lines and interaction vertices; one shared tree vertex, seven propagators, one loop.

K11 Swordfish on s-tree leg 1

`\STreeChannelCorr`

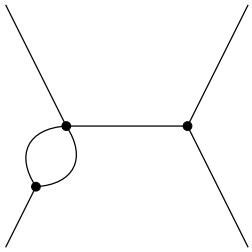


One trivalent loop vertex; two curves meet the actual quartic tree vertex.

One external insertion. No bridge or duplicate shared vertex.

K12 Swordfish on s-tree leg 2

`\STreeChannelCorr`

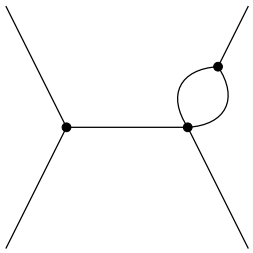


One trivalent loop vertex; two curves meet the actual quartic tree vertex.

One external insertion. No bridge or duplicate shared vertex.

K13 Swordfish on s-tree leg 3

`\STreeChannelCorr`

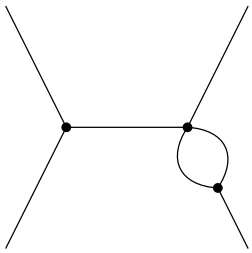


One trivalent loop vertex; two curves meet the actual quartic tree vertex.

One external insertion. No bridge or duplicate shared vertex.

K14 Swordfish on s-tree leg 4

`\STreeChannelCorr`



One trivalent loop vertex; two curves meet the actual quartic tree vertex.

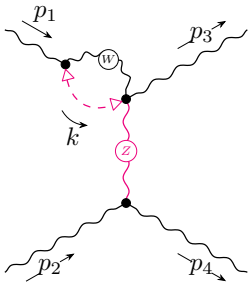
One external insertion. No bridge or duplicate shared vertex.

Independent loop styles at a shared interaction

Selected external momentum appears once; 1/2 incoming and 3/4 outgoing. The swordfish has no inner host segment.

K21 T-tree slot 1: mixed loop

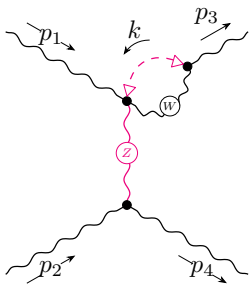
`\TTreeChannelCorr`



W loop, pink two-ended Z loop (k), pink Z exchange.
Species circles intentional; no central/W guide, caps or indices.

K22 T-tree slot 3: mixed loop

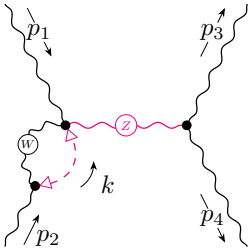
`\TTreeChannelCorr`



W loop, pink two-ended Z loop (k), pink Z exchange.
Species circles intentional; no central/W guide, caps or indices.

K23 S-tree slot 2: mixed loop

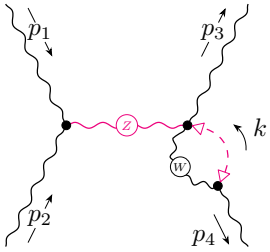
`\STreeChannelCorr`



W loop, pink two-ended Z loop (k), pink Z exchange.
Species circles intentional; no central/W guide, caps or indices.

K24 S-tree slot 4: mixed loop

`\STreeChannelCorr`



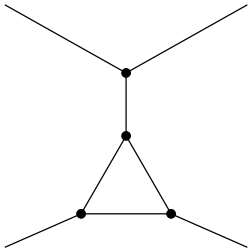
W loop, pink two-ended Z loop (k), pink Z exchange.
Species circles intentional; no central/W guide, caps or indices.

Regular tree-attached triangle corrections

Equilateral defaults; four trivalent interactions, one loop and one exchange bridge. Plain lines and no annotations.

L01 T-tree triangle: bottom

`\TTreeChannelCorr`

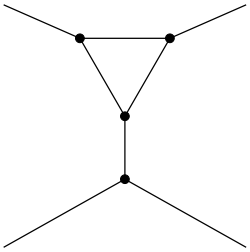


Three trivalent loop vertices; bridge to the other tree vertex.

One vertex correction. Equilateral defaults; explicit width allows other shapes.

L02 T-tree triangle: top

`\TTreeChannelCorr`

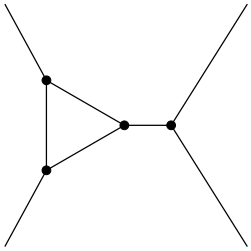


Three trivalent loop vertices; bridge to the other tree vertex.

One vertex correction. Equilateral defaults; explicit width allows other shapes.

L03 S-tree triangle: left

`\STreeChannelCorr`

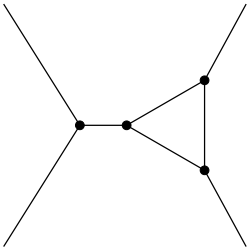


Three trivalent loop vertices; bridge to the other tree vertex.

One vertex correction. Equilateral defaults; explicit width allows other shapes.

L04 S-tree triangle: right

`\STreeChannelCorr`



Three trivalent loop vertices; bridge to the other tree vertex.

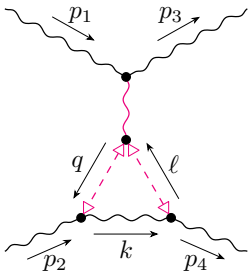
One vertex correction. Equilateral defaults; explicit width allows other shapes.

Independent lines and continuous momentum arrows

The same four graphs with mixed styles and outside labels. External 1/2 enter; 3/4 leave.

L11 T bottom: momentum and mixed lines

`\TTreeChannelCorr`

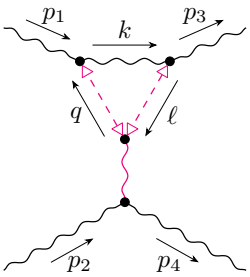


EW base and wavy exchange; pink two-ended loop edges.

No caps, circles, central guide or indices. Index polish remains D01.

L12 T top: momentum and mixed lines

`\TTreeChannelCorr`

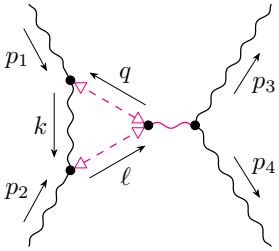


EW base and wavy exchange; pink two-ended loop edges.

No caps, circles, central guide or indices. Index polish remains D01.

L13 S left: momentum and mixed lines

`\STreeChannelCorr`

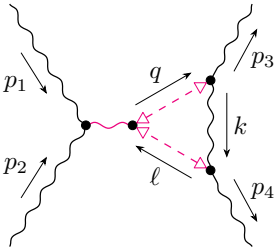


EW base and wavy exchange; pink two-ended loop edges.

No caps, circles, central guide or indices. Index polish remains D01.

L14 S right: momentum and mixed lines

`\STreeChannelCorr`



EW base and wavy exchange; pink two-ended loop edges.

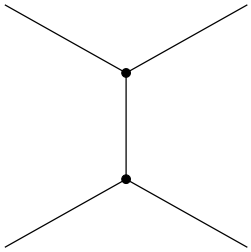
No caps, circles, central guide or indices. Index polish remains D01.

Consistent endpoints across trees and boxes

One default square outer frame; the large triangle fits without extending selected legs.

L21 Ordinary T tree

`\TTreeChannelCorr`

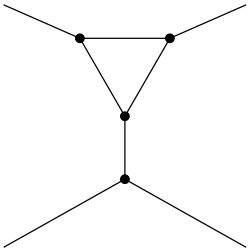


Common outer corners: x and y are both +/-2.10.

Comparison repeat; connectivity and inner geometry remain topology-specific.

L22 T triangle, same endpoints

`\TTreeChannelCorr`

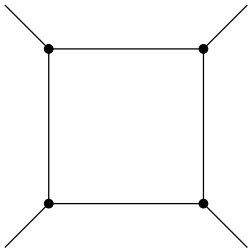


Common outer corners: x and y are both +/-2.10.

Comparison repeat; connectivity and inner geometry remain topology-specific.

L23 Box, same endpoints

`\BoxLoopCorr`

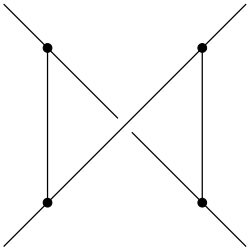


Common outer corners: x and y are both +/-2.10.

Comparison repeat; connectivity and inner geometry remain topology-specific.

L24 Crossed box, same endpoints

`\CrossBoxCorr`



Common outer corners: x and y are both +/-2.10.

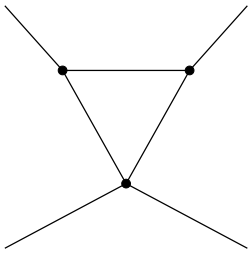
Comparison repeat; connectivity and inner geometry remain topology-specific.

The same endpoints in all triangle-contact orientations

These comparison repeats use the shared outer frame and retain their existing contact connectivity.

L31 Triangle-contact: bottom

`\TriangleContactCorr`

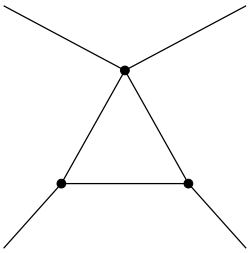


Same +/-2.10 outer corners as trees, box and crossed box.

Comparison repeat; the quartic vertex and internal spans are unchanged.

L32 Triangle-contact: top

`\TriangleContactCorr`

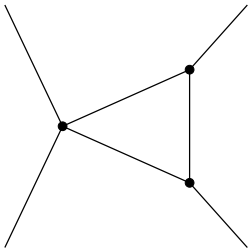


Same +/-2.10 outer corners as trees, box and crossed box.

Comparison repeat; the quartic vertex and internal spans are unchanged.

L33 Triangle-contact: left

`\TriangleContactCorr`

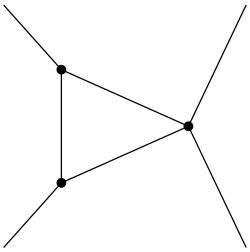


Same +/-2.10 outer corners as trees, box and crossed box.

Comparison repeat; the quartic vertex and internal spans are unchanged.

L34 Triangle-contact: right

`\TriangleContactCorr`



Same +/-2.10 outer corners as trees, box and crossed box.

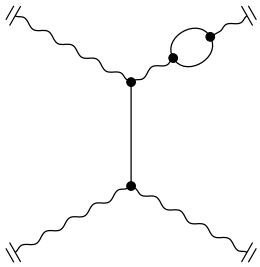
Comparison repeat; the quartic vertex and internal spans are unchanged.

External caps survive segment styling

The corrected external segment retains its logical leg cap. Internal insertion pieces stay uncapped.

N01 T bubble: double caps

`\TTreeChannelCorr`

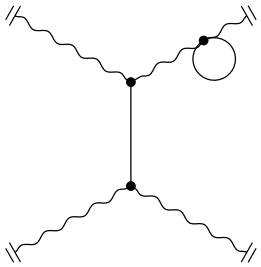


Four outer double caps retained.

Explicit segment caps override logical leg caps.

N02 T seagull: double caps

`\TTreeChannelCorr`

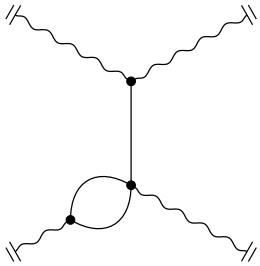


Four outer double caps retained.

Explicit segment caps override logical leg caps.

N03 T swordfish: double caps

`\TTreeChannelCorr`

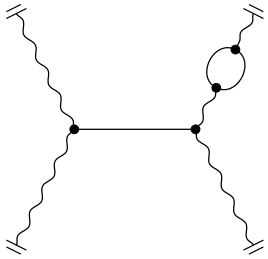


Four outer double caps retained.

Explicit segment caps override logical leg caps.

N04 S bubble: double caps

`\STreeChannelCorr`



Four outer double caps retained.

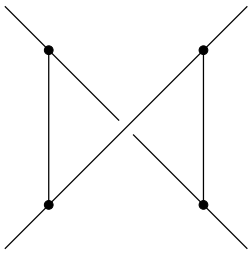
Explicit segment caps override logical leg caps.

Complementary internal crossings

Both layouts preserve the square corner frame and external slots. Plain lines expose the graph; the crossing is not a vertex.

P01 Crossed box: left/right fixed

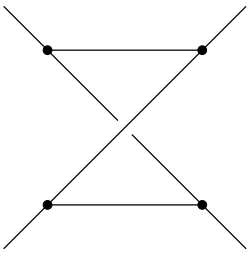
`\CrossBoxCorr`



Four trivalent vertices; the diagonal crossing is unconnected.
One split diagonal; no interaction at the crossing.

P02 Crossed box: top/bottom fixed

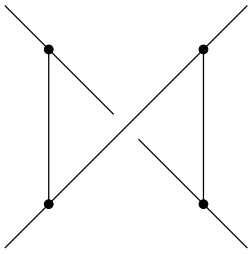
`\CrossBoxCorr`



Four trivalent vertices; the diagonal crossing is unconnected.
One split diagonal; no interaction at the crossing.

P03 Left/right: wider crossing gap

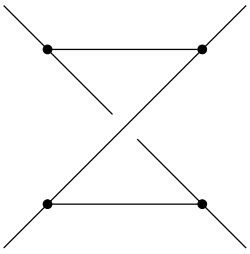
`\CrossBoxCorr`



Four trivalent vertices; the diagonal crossing is unconnected.
Same outer frame; the wider gap is a layout variant.

P04 Top/bottom: wider crossing gap

`\CrossBoxCorr`



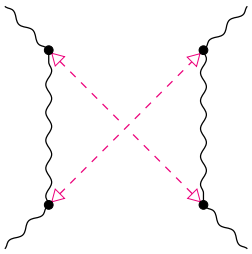
Four trivalent vertices; the diagonal crossing is unconnected.
Same outer frame; the wider gap is a layout variant.

Independent edges and annotations

The existing and complementary graphs share logical slots. These examples add only the named style or annotation feature.

P11 Left/right: mixed EW and Proca

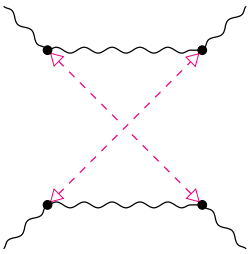
`\CrossBoxCorr`



Wavy EW fixed edges; pink Proca diagonals meet the real vertices.
No caps, circles, indices or momentum guides.

P12 Top/bottom: mixed EW and Proca

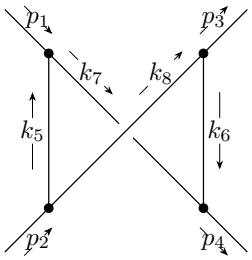
`\CrossBoxCorr`



Wavy EW fixed edges; pink Proca diagonals meet the real vertices.
No caps, circles, indices or momentum guides.

P13 Left/right: internal momentum gaps

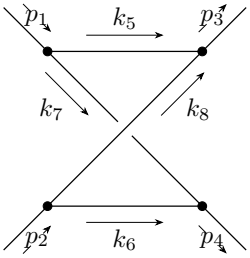
`\CrossBoxCorr`



Independent internal annotations; labels sit in measured arrow gaps.
Indices remain a separate D01 appearance review.

P14 Top/bottom: continuous momentum arrows

`\CrossBoxCorr`



Continuous arrows with labels beside their logical edges.
Per-edge directions and positions remain independently adjustable.